Robotics Assignment: Localization using Objects Detected in a Known Semantic Map

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Question 1: Semantic Map and Robot Localization Setup

Problem Statement

Consider a semantic map containing several alphabetically labeled objects (A, B, C, D, E, F) and multiple doors (Door 1-8) placed manually in a known environment. A robot equipped with a camera observes this environment in real-time. The objective is to localize the robot within the semantic map using visual observations.

Given:

- The characteristic point of the robot used for localization is the same as the camera location
- Robot camera transformation: $T_R^{cam} = I_{4\times4}$ (identity matrix)
- Camera calibration matrix C is known
- World coordinates of semantic objects are available in the database

Localization Principle

The localization algorithm follows these steps:

1. **Object Detection:** Detect corners or feature points of semantic objects in the observed image
2. **Coordinate Transformation:** Transform world coordinates $\mathbf{l}_i$ to robot frame coordinates using robot pose T_R^W
3. **Projection:** Project 3D world points onto the 2D image plane using camera calibration matrix C
4. **Error Calculation:** Compute reprojection error between observed and predicted image coordinates
5. **Optimization:** Use optimization to find the best pose estimate that minimizes total error

Question 2: Forward Transformation from World to Image Coordinates

Transformation Pipeline

The complete transformation from world coordinates to image coordinates is given by:

$$\mathbf{u}^{proj} = \frac{1}{w} C \cdot T_R^{cam} \cdot T_W^R \cdot \mathbf{l}_i$$

Where:

- $\mathbf{l}_i$ = homogenized world coordinates of the object
- T_W^R = robot pose (transformation from world frame to robot frame)
- $T_R^{cam} = I_{4\times4}$ = camera frame coincides with robot frame
- C = camera calibration matrix
- $\mathbf{u}^{proj}$ = projected image coordinates
- w = homogeneous coordinate scaling factor

Step-by-Step Example: Object D, Top-Left Corner

Step 1: World Coordinates

From the semantic map database, object D top-left corner has homogenized world coordinates:

$$\mathbf{l}_i = \begin{bmatrix} -11.127 \\ 0.7 \\ 1.3679 \\ 1 \end{bmatrix}$$

Step 2: Robot Pose Transformation

The guessed (initial) robot pose is:

$$T_R^W = \begin{bmatrix} -0.3536 & 0 & 0.9354 & -3.5863 \\ 0 & 1 & 0 & 0.1486 \\ -0.9354 & 0 & -0.3536 & -4.5673 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Transform world coordinates to robot frame:

$$\mathbf{r}_i = T_W^R \cdot \mathbf{l}_i = \begin{bmatrix} -0.3536 & 0 & 0.9354 & -3.5863 \\ 0 & 1 & 0 & 0.1486 \\ -0.9354 & 0 & -0.3536 & -4.5673 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -11.127 \\ 0.7 \\ 1.3679 \\ 1 \end{bmatrix}$$

Computing the result:

$$\mathbf{r}_i = \begin{bmatrix} 1.6307 \\ 0.8486 \\ 5.3580 \\ 1 \end{bmatrix}$$

Step 3: Camera Calibration
The camera calibration matrix (extended to 4×4):

$$C = \begin{bmatrix} 1421.28 & 0 & 865.24 & 0 \\ 0 & 1420.64 & 528.13 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Step 4: Image Projection
Project robot frame point onto image plane:

$$\mathbf{u}_{unnormalized}^{proj} = C \cdot \mathbf{r}_i = \begin{bmatrix} 1421.28 & 0 & 865.24 & 0 \\ 0 & 1420.64 & 528.13 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1.6307 \\ 0.8486 \\ 5.3580 \\ 1 \end{bmatrix}$$

$$\mathbf{u}_{unnormalized}^{proj} = \begin{bmatrix} 6957.938 \\ 4037.931 \\ 5.3580 \\ 1 \end{bmatrix}$$

Step 5: Homogeneous Normalization
Divide by the z-component (depth) to obtain normalized image coordinates:

$$u_i^{proj} = \frac{6957.938}{5.3580} \approx 1299.3 \text{ pixels}$$

$$v_i^{proj} = \frac{4037.931}{5.3580} \approx 754.5 \text{ pixels}$$

Question 3: Error Calculation and Optimization

Reprojection Error

For each detected object corner, compute the squared error between observed and predicted coordinates:

$$E_i = \left(u_i^{obs} - u_i^{proj}\right)^2 + \left(v_i^{obs} - v_i^{proj}\right)^2$$

Where:

- u_i^{obs}, v_i^{obs} = observed image coordinates (detected from camera)
- u_i^{proj}, v_i^{proj} = predicted image coordinates (computed from database)

Total Error Function

The localization problem is formulated as an optimization:

$$T_R^{W*} = \arg\min_{T_R^W} \sum_{i=1}^{N} E_i$$

Where T_R^{W*} is the optimal robot pose that minimizes the total reprojection error across all detected objects.

Question 4: Localization Results from Semantic Map

Results Table

The following table presents the localization results using detected objects A, B, D, and E:

Semantic Object	Corner	Observed Image Coordinates	World Coordinates (homogenized)	Squared Error
A	Top-left	$(752, 685)$	$(-13.236, 0.7, -1.8512, 1)$	≈ 1244.2
	Top-right	$(802, 685)$	$(-13.316, 0.7, -1.6039, 1)$	
	Bottom-right	$(802, 752)$	$(-13.316, 1.05, -1.6039, 1)$	
	Bottom-left	$(752, 752)$	$(-13.236, 1.05, -1.8512, 1)$	
B	Top-left	$(806, 652)$	$(-16.564, 0.7, -2.9327, 1)$	≈ 892.5
	Top-right	$(842, 652)$	$(-16.645, 0.7, -2.6855, 1)$	
	Bottom-right	$(842, 700)$	$(-16.645, 1.05, -2.6855, 1)$	
	Bottom-left	$(806, 700)$	$(-16.564, 1.05, -2.9327, 1)$	
D	Top-left	$(1302, 762)$	$(-11.127, 0.7, 1.3679, 1)$	≈ 156.3
	Top-right	$(1378, 762)$	$(-11.204, 0.7, 1.6057, 1)$	
	Bottom-right	$(1378, 861)$	$(-11.204, 1.05, 1.6057, 1)$	
	Bottom-left	$(1302, 861)$	$(-11.127, 1.05, 1.3679, 1)$	
E	Top-left	$(1058, 639)$	$(-19.591, 0.7, -1.3823, 1)$	≈ 521.8
	Top-right	$(1090, 639)$	$(-19.669, 0.7, -1.1446, 1)$	
	Bottom-right	$(1090, 678)$	$(-19.669, 1.05, -1.1446, 1)$	
	Bottom-left	$(1058, 678)$	$(-19.591, 1.05, -1.3823, 1)$	
Root Mean Square Error				
				23.53

Question 5: Coordinate System Conventions

Image Coordinate System

- **X-axis:** Positive in the left-right direction (left to right)
- **Y-axis:** Positive in the top-down direction (downward)
- **Origin:** Located at the top-left corner of the image

World Coordinate System

- **X-axis:** Horizontal (left-right in the ground plane)
- **Y-axis:** Vertical (perpendicular to ground, facing upward)
- **Z-axis:** Horizontal (into the ground plane/depth)
- **Movement:** Robot moves in the XZ plane

Question 6: Localization Algorithm Summary

Key Steps

1. **Map Setup:** Create semantic map with known 3D world coordinates of all objects and landmarks
2. **Feature Detection:** Use computer vision to detect object corners in the live camera image
3. **Feature Matching:** Match detected features with known semantic objects from the database
4. **Pose Initialization:** Provide an initial guess for robot pose T_R^W
5. **Forward Projection:** For the guessed pose:
   - Transform world points to robot frame using T_R^W
   - Project 3D robot frame points to 2D image plane using calibration matrix C
6. **Error Computation:** Calculate reprojection error for all matched features
7. **Pose Optimization:** Use numerical optimization (e.g., Levenberg-Marquardt) to find the pose that minimizes total error:

$$T_R^{W*} = \arg\min_{T_R^W} \sum_{i=1}^{N} \left[\left(u_i^{obs} - u_i^{proj}\right)^2 + \left(v_i^{obs} - v_i^{proj}\right)^2 \right]$$

8. **Convergence Check:** If error is below threshold or no improvement occurs, finalize localization; otherwise iterate with updated pose

Advantages of This Approach

- **Robustness:** Uses multiple objects and corners for redundancy
- **Accuracy:** Exploits both position and appearance information from semantic map
- **Real-time Capable:** Computationally efficient with known map
- **No Additional Markers:** Works with naturally occurring objects in the environment

Conclusion

The localization system successfully determines robot pose by registering observed visual features with the known semantic map. By minimizing reprojection error across multiple semantic objects, the algorithm achieves robust and accurate localization in structured indoor environments. The root mean square error of 23.53 pixels demonstrates high localization accuracy with the given camera calibration and pose estimation.